

Experiment:- Study of LAN Topology, Network devices and Address configuration.

Objective:

- 1.) To study the LAN topology used in College Campus.
- 2.) To identify networking devices & understand their function.
- 3.) To analyse wired & wireless communication method used in the network.
- 4.) To measure speed of network for wired & wireless connections.
- 5.) To obtain MAC, IP Address & subnet mask.

Problem Statement - 1

- Draw & identify LAN Topology of College Campus and clearly label all components such as router, switches & etc.

Solution:

The college Campus network observed this study consists of:

- Layer-3 core switches in the server room.
- Managed by POE + switches.
- Wireless Access Points (APs)
- Ethernet port in classroom & laboratories.

Networking Topology

Hybrid Hierarchical Topology which combines:

- 1.) Tree Topology
- 2.) Extended star Topology

3> Ring Topology (Server Room)

The network hierarchical layer

- ① Core Layer
- ② Distribution Layer
- ③ Access Layer

Core Layer

- Core Layer contains server room and server room and server room have Firewall, Router, 2-13+ Switches, Patch Rack, Server, inverter etc.

Distribution Layer

- Distribution layer contains 4-switches.

- The 4 switches are —

- ① Main Build → R7 (Room no. - 213, 2nd floor)
- ② Main Build → R1 (Room no. - 301, 3rd floor)
- ③ Main Build → R6 (Room no. - 113, 1st floor)
- ④ Main Build → R2 (Room no. - 201, 2nd floor)

* This 4-switches contains →

- ① 18 Port Gigabit Easy Smart Switch with 16 port.

TL-SG1218MPE

- ② SG 3428 XMP → 24 port Gigabit & 4 port 10 GE SFP+, 12+ Managed switches with 24 Port POE+.

→ Now, let's discuss these 2 above components

① TP-Link TL-SG1218MPE

- Likely used for : CCTV, camera, Wifi access point, etc
- 18 Port Gigabit "Easy Smart" switch.
- 16 port support POE+.

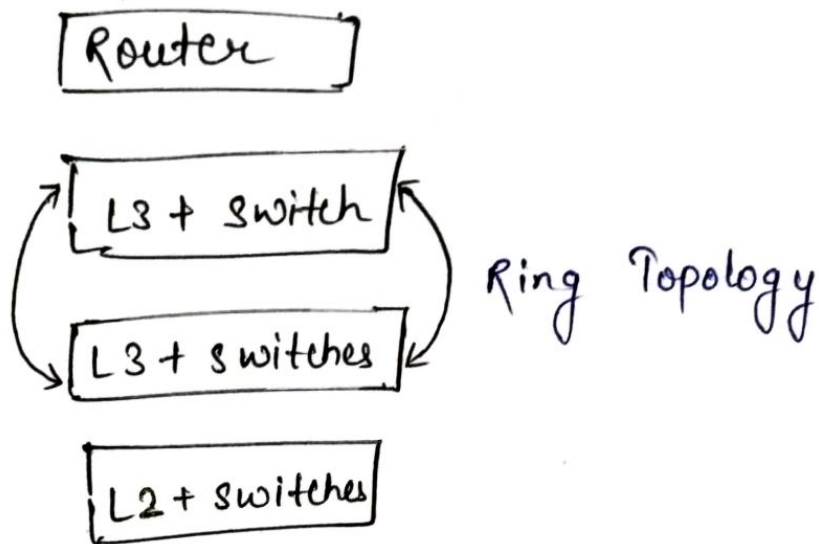
② TP-Link Omada SG3428XMP

- Higher-end managed layer 2+ switches.
- 24 POE+ port, Uplinks, 10G SFP.
- Usually acts as a main distribution for each floor.

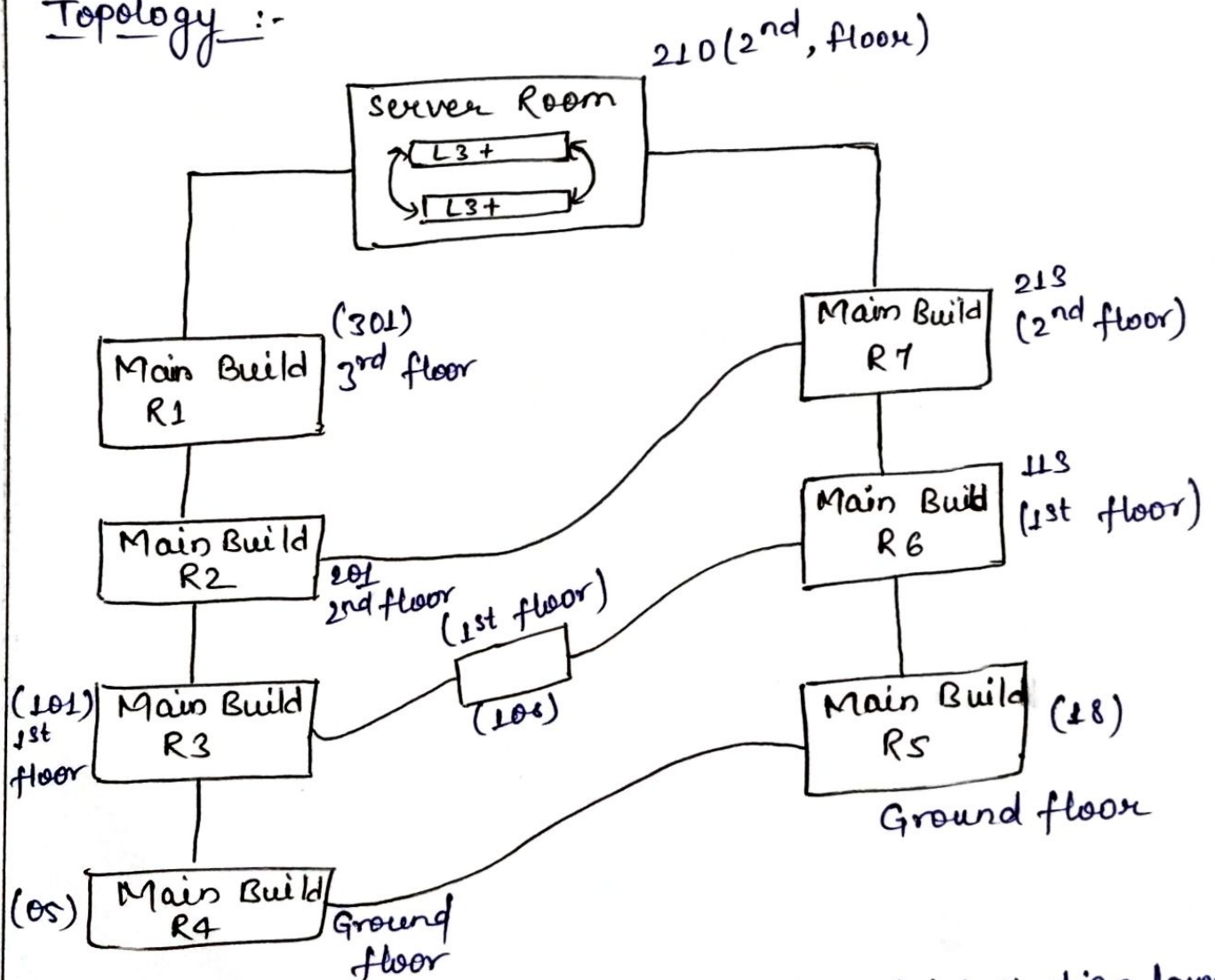
→ Access Layer:

- Access Layer contain 3-switches i.e
 - 1> Main Build R1 - [Room no. - 301, 3rd floor]
 - 2> Main Build R4 - [Room no. - 05, Ground Floor]
 - 3> Main Build R5 - [Room no. - 18, Ground Floor]
- Every switches acts as access layer helps to provide network connectivity to all classroom, lab, faculty room etc.
- Access Layer directly connect:
 - ① Ethernet Port
 - ② Wireless AP
 - ③ Lab System
 - ④ Outdoor Wireless AP (ex-018)

→ Server Room Architecture / Topology.



Topology :-

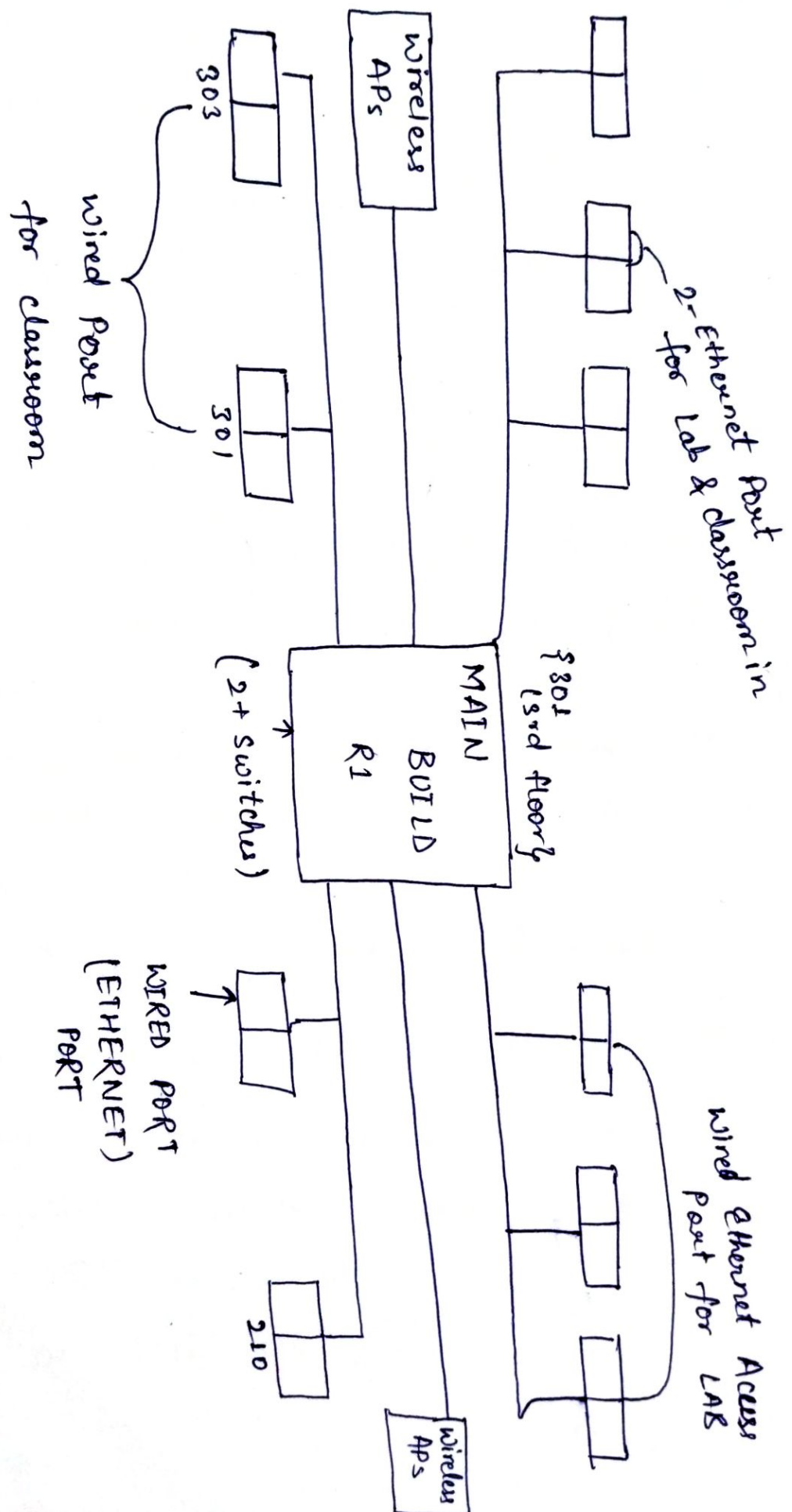


→ R1, R7, R2, R6 → Distribution layer

→ Server Room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer

Access Layer Switch (3rd floor - 301)





Performance

Run new task

CPU
25% 1.35 GHz

Memory
9.1/15.7 GB (58%)

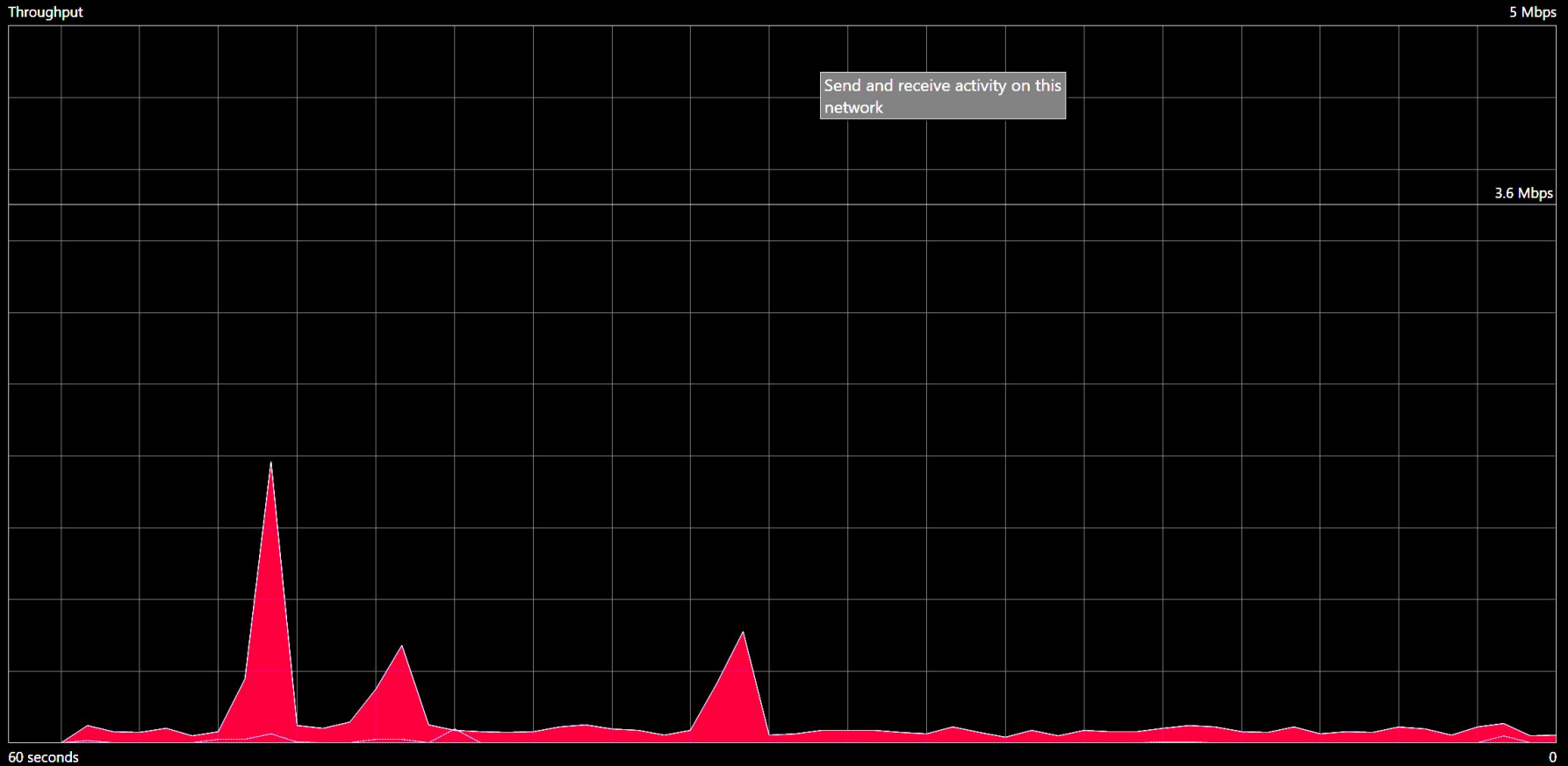
Disk 0 (C: D:)
SSD (NVMe)
1%

Wi-Fi
Wi-Fi
S: 0 R: 56.0 Kbps

GPU 0
Intel(R) Iris(R) Xe Graphi
2%

Wi-Fi

Killer(R) Wi-Fi 6E AX1675i 160MHz Wireless Network Adapter (211NGW)



Send
0 Kbps

Receive
56.0 Kbps

Adapter name: Wi-Fi

SSID: GECK-Campus-WiFi

Connection type: 802.11ac

IPv4 address: 10.85.23.225

IPv6 address: fe80::f1aa:b406:7940:9fe5%14

Signal strength:

Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\deeps> netsh wlan

The following commands are available:

Commands in this context:

?	- Displays a list of commands.
add	- Adds a configuration entry to a table.
connect	- Connects to a wireless network.
delete	- Deletes a configuration entry from a table.
disconnect	- Disconnects from a wireless network.
dump	- Displays a configuration script.
export	- Saves WLAN profiles to XML files.
help	- Displays a list of commands.
IHV	- Commands for IHV logging.
refresh	- Refresh hosted network settings.
reportissues	- Generate WLAN smart trace report.
set	- Sets configuration information.
show	- Displays information.
start	- Start hosted network.
stop	- Stop hosted network.

To view help for a command, type the command, followed by a space, and then type ?.

```
PS C:\Users\deeps> netsh wlan show interface
```

There is 1 interface on the system:

Name	: Wi-Fi
Description	: Killer(R) Wi-Fi 6E AX1675i 160MHz Wireless Network Adapter (211NGW)
GUID	: bbcb55de-c354-40d5-bbf6-56e3eb86cccf
Physical address	: 10:91:d1:ed:d1:81
Interface type	: Primary
State	: connected
SSID	: GECK-Campus-WiFi
AP BSSID	: b0:19:21:74:07:b9
Band	: 5 GHz
Channel	: 48
Connected Akm-cipher	: [akm = 00-0f-ac:02, cipher = 00-0f-ac:04]
Network type	: Infrastructure
Radio type	: 802.11ac
Authentication	: WPA2-Personal
Cipher	: CCMP
Connection mode	: Auto Connect
Receive rate (Mbps)	: 390
Transmit rate (Mbps)	: 390
Signal	: 72%
Rssi	: -67
Profile	: GECK-Campus-WiFi
QoS MSCS Configured	: 0
QoS Map Configured	: 0
QoS Map Allowed by Policy	: 0

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::f1aa:b406:7940:9fe5%14
IPv4 Address. : 10.85.23.225
Subnet Mask : 255.255.248.0
Default Gateway : 10.85.16.1

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :

PS C:\Users\deeps> getmac

Physical Address	Transport Name
10-91-D1-ED-D1-81	\Device\Tcpip_{BBCB55DE-C354-40D5-BBF6-56E3EB86CCCF}
10-91-D1-ED-D1-85	Media disconnected

PS C:\Users\deeps>

Difference between Wired & Wireless speed?

Wired Network

- Faster Communication
- Stable Connection
- Low latency
- Dedicated physical medium

Wireless Network

- Slower Communication
- Affected by distance/Interference
- Higher latency
- Shared radio Medium.

- Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.
- Wireless connections ~~uses~~ use radio waves which are affected by interference, distance obstacles multiple connected users.

Q) Role of Subnet mask in Network?

A Subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

It is important for:

- ① Network Segment
- ② Efficient Routing
- ③ Communication b/w devices.

OBSERVATION

During the study of the college network infrastructure

- ① A centralized server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-Link POE+ switches were used in network cabinets.

Advantages of Existing Network

- Scalable architecture.
- Centralized management
- supports POE+ device.
- Easy expansion of floors & classroom.
- Support both wireless & wired.
- Redundant link for failure.

CONCLUSION

The observed campus LAN follows a hybrid hierarchical topology consists of Tree, Extended star, Ring. The network uses managed POE switches, wireless access point, Ethernet port. & centralized server infrastructure to provide efficient wired & wireless communication across the campus.